

# Differentially Private Data Publishing and Analysis: A Survey

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**Abstract**—Differential privacy is an essential and prevalent privacy model that has been widely explored in recent decades. This survey provides a comprehensive and structured overview of two research directions: differentially private data publishing and differentially private data analysis. We compare the diverse release mechanisms of differentially private data publishing given a variety of input data in terms of query type, the maximum number of queries, efficiency, and accuracy. We identify two basic frameworks for differentially private data analysis and list the typical algorithms used within each framework. The results are compared and discussed based on output accuracy and efficiency. Further, we propose several possible directions for future research and possible applications.

**Index Terms**—Differential privacy, privacy preserving data publishing, privacy preserving data analysis

## 1 INTRODUCTION

OVER the past two decades, digital information collected by corporations, organizations, and governments has resulted in a vast number of datasets, and the speed of such data collection has increased dramatically over the last few years. Typically, a data collector, also known as a *curator*, is in charge of publishing data for further analysis [1]. However, most collected datasets contain private or sensitive information. Even though curators can apply several simple anonymization techniques, sensitive personal information still has a high probability of being disclosed [2]. Privacy preservation has, therefore, become an urgent issue that needs to be addressed.

Fig. 1 shows how a trusted curator preserves an original dataset that includes personal information. The curator provides aggregated information to public users, who may use this information for further investigation. We divide this process into *data publishing* and *data analysis* based on the purpose of its release. Data publishing aims to share datasets or some query results to the public. In some literature, this scenario is known as data sharing or data release. Another scenario, where a curator provides data models directly to the public, is normally defined as data analysis. The shared models may be associated with particular algorithms, such as data mining or machine learning algorithms.

Research communities have proposed various methods to preserve individual privacy in these two scenarios. The methods and their privacy criteria are defined as a *privacy model*. As shown in Fig. 1, a privacy model sits between a

trusted curator and untrusted public users. *Differential privacy* is one such new and promising privacy model. It ensures that the ability of an adversary to inflict harm on any individual in a dataset is essentially the same, independent of whether any individual opts into, or out of, the dataset [3]. Compared to previous privacy models, differential privacy can successfully resist most of privacy attacks and provide a provable privacy guarantee.

Interest in differential privacy is very high and its central notion spans a range of research areas, from the privacy community to areas of data science such as machine learning, data mining, statistics, and learning theory. Much work has also been conducted in a number of application domains, including social networks, location privacy, and recommender systems. Fig. 2 shows some of these key elements to be further discussed in this survey.

### 1.1 Outline and Survey Overview

The initial work on differential privacy was pioneered by Dwork et al. [3] in 2006. Over the last decade, several surveys on differential privacy have been completed:

- 1) The first survey by Dwork et al. [4] recalled the definition of differential privacy and two of its principle mechanisms with the aim of showing how to apply these techniques in data publishing.
- 2) Dwork et al.'s report [5] exploited the difficulties that arise when data publishing encounters prospective solutions in the context of statistics analysis. It identified several research issues in data analysis that had not been adequately investigated at that time.
- 3) In a review [6], Dwork et al. provided an overview of the principal motivating scenarios, together with a summary of future research directions.
- 4) Sarwate et al. [7] focused on privacy preserving for continuous data to solve the problems in signal processing.
- 5) A book by Dwork et al. [8] presented an accessible starting place for anyone looking to learn about the theory of differential privacy.

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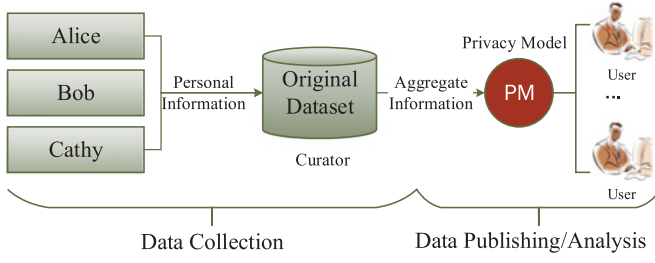


Fig. 1. Privacy model.

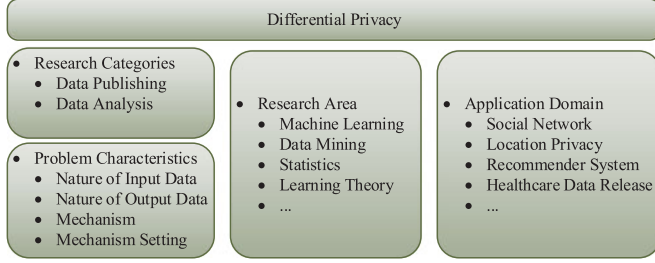


Fig. 2. Key elements associated with differential privacy.

These surveys and the book focus on the concepts and theories of differential privacy; however, the mathematical theories are not easily implemented into applications directly. Yet, after more than ten years of theoretical development, a significant number of new technologies and applications have appeared in this area. We believe that now is a good time to summarize the new technologies and address the gap between theory and application.

Here we attempt to find a clearer way to present the concepts and practical aspects of differential privacy for the data mining research community.

- We avoid detailed theoretical analysis of related differentially private algorithms and instead place more focus on its practical aspects, which may benefit applications in the real world.
- We try to avoid repeating many references that have already been analyzed extensively in the above well-cited surveys.
- Even though differential privacy covers multiple research directions, we restrict our observations to data publishing and data analysis scenarios, which are the most popular scenarios in the research community.

Table 1 defines the scope of these two research directions within the survey. Mechanism design for data publishing is normally independent from its publishing targets, as the goal of publishing is to release query answers or a dataset for further use and, hence, is unknown to the curator. The mechanism design for data analysis aims to preserve privacy during the analysis process. The curator already knows the details of the analysis algorithm, so the mechanism is associated with the analysis algorithm.

## 2 DIFFERENTIAL PRIVACY PRELIMINARIES

### 2.1 Notation

We consider a finite data universe  $\mathcal{X}$  with the size  $|\mathcal{X}|$ . Let  $r$  represent a record with  $d$  attributes; a dataset  $D$  is an unordered set of  $n$  records sampled from the universe  $\mathcal{X}$ . Two datasets  $D$  and  $D'$  are defined as neighboring datasets if

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TABLE 1  
Comparison between Differentially Private Data Publishing and Analysis

|           | Differentially private data publishing | Differentially private data analysis   |
|-----------|----------------------------------------|----------------------------------------|
| Mechanism | independent mechanism                  | coupled with a particular algorithm    |
| Input     | various data types                     | transaction dataset (training samples) |
| Output    | query answers or datasets              | various models                         |

TABLE 2  
Notations

| Notations     | Explanation          | Notations       | Explanation        |
|---------------|----------------------|-----------------|--------------------|
| $\mathcal{X}$ | Universe             | $D$             | Dataset            |
| $D'$          | Neighbour dataset    | $\mathcal{D}$   | Data distribution  |
| $r$           | Record               | $d$             | Dataset dimension  |
| $n$           | Size of dataset      | $N$             | size of histogram  |
| $f$           | Query                | $F$             | Query set          |
| $m$           | Query number         | $\mathcal{M}$   | Mechanism          |
| $\hat{f}$     | Noisy output         | $\eta$          | Noise              |
| $\epsilon$    | Privacy budget       | $\Delta f$      | Sensitivity        |
| $G$           | Graph data           | $t, T$          | Time sequence      |
| $\mathbf{w}$  | Output model         | $VC(\cdot)$     | VC dimension       |
| $c$           | Concept              | $\mathcal{C}$   | Concept set        |
| $h$           | Hypothesis           | $H$             | Hypothesis set     |
| $\ell(\cdot)$ | Loss function        | $\theta$        | Threshold          |
| $\delta$      | Confidence parameter | $\alpha, \beta$ | Accuracy parameter |

they differ in one record. A query  $f$  is a function that maps dataset  $D$  to an abstract range  $\mathcal{R}$ :  $f : D \rightarrow \mathcal{R}$ . A group of queries is denoted as  $F$ . We use the symbol  $m$  to represent the number of queries in  $F$ .

The aim of differential privacy is to mask the differences in query  $f$  between neighboring datasets. The maximal difference in the results of query  $f$  is defined as the *sensitivity*  $\Delta f$ . Differential privacy is generally achieved by a mechanism  $\mathcal{M}$  which is a randomized algorithm that accesses the database and implements some functionality. Randomized output is denoted by a circumflex over the notation. For example,  $\hat{f}(D)$  denotes the randomized answer of querying  $f$  on  $D$ . Table 2 summarizes the notations used in the following sections.

### 2.2 Differential Privacy

**Definition 1 (( $\epsilon, \delta$ )-differential privacy).** [9] A randomized mechanism  $\mathcal{M}$  gives ( $\epsilon, \delta$ )-differential privacy for every set of outputs  $\Omega$ , and for any neighbouring datasets of  $D$  and  $D'$ , if  $\mathcal{M}$  satisfies

$$Pr[\mathcal{M}(D) \in \Omega] \leq \exp(\epsilon) \cdot Pr[\mathcal{M}(D') \in \Omega] + \delta. \quad (1)$$

If  $\delta = 0$ , the randomized mechanism  $\mathcal{M}$  gives  $\epsilon$ -differential privacy by its strictest definition. ( $\epsilon, \delta$ )-differential privacy provides freedom to violate strict differential privacy for some low probability events.  $\epsilon$ -differential privacy is usually called *pure differential privacy*, while ( $\epsilon, \delta$ )-differential privacy with  $\delta > 0$  is called *approximate differential privacy* [10].

### 2.3 The Privacy Budget Composition

In Definition 1, the parameter  $\epsilon$  refers to the *privacy budget* [9], which controls the level of privacy guarantee achieved by mechanism  $\mathcal{M}$ . A smaller  $\epsilon$  represents a

stronger privacy level. Two privacy budget compositions theorems are widely used: *sequential composition* [11] and *parallel composition* [12].

**Theorem 1 (Parallel Composition).** Suppose we have a set of privacy mechanisms  $\mathcal{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_m\}$ . If each  $\mathcal{M}_i$  provides a  $\epsilon_i$ -differential privacy guarantee on a disjointed subset of the entire dataset,  $\mathcal{M}$  will provide  $(\max\{\epsilon_1, \dots, \epsilon_m\})$ -differential privacy.

**Theorem 2 (Sequential Composition).** Suppose a set of privacy mechanisms  $\mathcal{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_m\}$  are sequentially performed on a dataset, and each  $\mathcal{M}_i$  provides  $\epsilon$ -differential privacy guarantee,  $\mathcal{M}$  will provide  $(m \cdot \epsilon)$ -differential privacy.

## 2.4 The Sensitivity

*Sensitivity* is the parameter that determines how much perturbation is required for a particular query in a mechanism.

**Definition 2 (Sensitivity).** [6] For a query  $f : D \rightarrow \mathcal{R}$ , and neighboring datasets  $D$  and  $D'$ , the sensitivity of  $f$  is defined as

$$\Delta f = \max_{D, D'} \|f(D) - f(D')\|_1. \quad (2)$$

Sensitivity  $\Delta f$  is only related to the type of query  $f$ . It considers the maximal difference between the query results on neighboring datasets.

## 2.5 The Principle Differential Privacy Mechanisms

Any mechanism meeting Definition 1 can be considered as differentially private. Currently, two basic mechanisms are widely used to guarantee differential privacy: the Laplace mechanism [13] and the exponential mechanism [11].

### 2.5.1 The Laplace Mechanism

The Laplace mechanism adds independent noise to the true answer. We use  $Lap(b)$  to represent the noise sampled from a Laplace distribution with a scaling of  $b$ .

**Definition 3 (Laplace mechanism).** [6] For a function  $f : D \rightarrow \mathcal{R}$  over a dataset  $D$ , the mechanism  $\mathcal{M}$  in Eq. (3) provides the  $\epsilon$ -differential privacy

$$\mathcal{M}(D) = f(D) + Lap\left(\frac{\Delta f}{\epsilon}\right). \quad (3)$$

### 2.5.2 The Exponential Mechanism

For non-numeric queries, differential privacy uses an exponential mechanism to randomize the results, and this is paired with a *score function*  $q(D, \phi)$  that evaluates the quality of an output  $\phi$ . Defining a score function is application-dependent and different applications lead to various score functions.

**Definition 4 (Exponential mechanism).** [11] Let  $q(D, \phi)$  be a score function of dataset  $D$  that measures the quality of output  $\phi \in \Phi$ ,  $\Delta q$  represents the sensitivity of  $\phi$ . The exponential mechanism  $\mathcal{M}$  satisfies  $\epsilon$ -differential privacy if

$$\mathcal{M}(D) = \left( \text{return } \phi \propto \exp\left(\frac{\epsilon q(D, \phi)}{2\Delta q}\right) \right). \quad (4)$$

## 2.6 Utility Measurement of Differential Privacy

Several utility measurements are used in both data publishing and analysis when the privacy level is fixed to  $\epsilon$ .

- *Noise size measurement*: the easiest approach is to calibrate how much noise is added to the query results. A smaller amount of noise indicates a higher utility. This utility measurement has been widely used in data publishing.
- *Error measurement*: utility can be evaluated by the difference between the non-private output and the private output. The error measurement is normally represented by a bound with accuracy parameters [14]:

**Definition 5 ( $(\alpha, \beta)$ -useful).** A mechanism  $\mathcal{M}$  is  $(\alpha, \beta)$ -useful if

$$\Pr\left(\max_{f \in F} |F(D) - \hat{F}(D)| \leq \alpha\right) > 1 - \beta, \quad (5)$$

where  $\alpha$  is the accuracy parameter that bounds the error. For different publishing scenarios, the error measurement can be interpreted in various ways. For synthetic dataset publishing, Eq. (5) can be interpreted as

$$\Pr\left(\max_{f \in F} |F(D) - F(\hat{D})| \leq \alpha\right) > 1 - \beta. \quad (6)$$

For data analysis, the utility measurement normally depends on the analysis algorithms. Suppose the algorithm is denoted by  $\mathcal{M}$  and the private algorithm is denoted by  $\hat{\mathcal{M}}$ , Eq. (5) can be interpreted as

$$\Pr(|\mathcal{M}(D) - \hat{\mathcal{M}}(D)| \leq \alpha) > 1 - \beta. \quad (7)$$

Eq. (7) has several implementations in data analysis, such as risk bound and sample complexity, which are presented in Section 4.

## 3 DIFFERENTIALLY PRIVATE DATA PUBLISHING

Differentially private data publishing aims to output aggregate information to the public without disclosing any individual's record. This problem can be presented as follows: if a curator has a dataset  $D$  and receives a query set  $F = \{f_1, \dots, f_m\}$ , they are required to answer each query  $f_i \in F$  subject to the constraints of differential privacy.

Two settings, interactive and non-interactive, are involved in this publishing scenario. In the interactive setting, a query  $f_i$  can not be issued until the answer to the previous query  $f_{i-1}$  has been published. In the non-interactive setting, all queries are given to the curator at one time, and the curator can provide answers with full knowledge of the query set.

An example to show the difference between the two settings is presented in Table 3. Queries to the curator may presented as follows:

- $f_1$ : How many patients have diabetes at the age of 40 – 79?
- $f_2$ : How many patients have diabetes at the age of 40 – 59?

Suppose the privacy budget  $\epsilon$  is fixed for each query. In the interactive setting, the curator will first get  $f_1$ , then count the number of patients who have diabetes between the ages of 40 and 79 and add independent Laplace noise to



TABLE 3  
Medical Table

| Name  | Age | Diabetes |
|-------|-----|----------|
| Alen  | 25  | N        |
| Bob   | 29  | N        |
| Cathy | 35  | Y        |
| David | 41  | Y        |
| Emily | 56  | N        |
| ...   | ... | ...      |
| Emma  | 21  | Y        |

TABLE 4  
Frequency Table

| Age   | No. of patients with diabetes | Variable |
|-------|-------------------------------|----------|
| 60-79 | 41                            | $x_1$    |
| 40-59 | 32                            | $x_2$    |
| 20-39 | 8                             | $x_3$    |
| 0-19  | 1                             | $x_4$    |

the number with a sensitivity equal to 1,  $Lap(1/\epsilon)$ . When  $f_2$  is then submitted to the curator,  $f_2$  will be answered with sensitivity equal to 2, as changing one person in the table may change the results of both queries. The total noise added to the query set is  $Lap(1/\epsilon) + Lap(2/\epsilon)$ .

In non-interactive settings, both queries are submitted to the curator at the same time. The sensitivity measured for both queries is 2. The total noise added to the query set is  $2 * Lap(2/\epsilon)$ , which is larger than the interactive setting. The correlation between queries leads to a higher sensitivity. Therefore, non-interactive settings normally incur more noise than interactive settings.

The above example presents the differences between the two settings and shows that the amount of noise increases dramatically when queries are correlated to each other. In addition, for a dataset with size  $n$ , the Laplace mechanism can only answer, at most, sub-linear in  $n$  number of queries to a certain level of accuracy [15].

These weaknesses make the Laplace mechanism impractical in the scenarios that require answering large amounts of queries. New mechanisms are required. Table 5 summarizes the problem characteristics of differentially private data publishing, in which mechanism design focuses on the number of queries, the accuracy of the output, and the computational efficiency.

### 3.1 Publishing Mechanisms

We categorize existing mechanisms into several types: transformation, dataset partitioning, query separation and iteration. Tables 3 and 4 are again used to show the key ideas in each type.

- *Transformation*: Transformation mechanisms map the original dataset to a new structure to adjust the sensitivity or level of noise. In the above example, the original dataset can be transferred to a frequency dataset as shown in Table 4.

In the new structure,  $f_2$  can be answered directly by the second row with a sensitivity equal to 1.  $f_1$  can be answered by combining  $\hat{f}_2$  and the noisy result of first row of Table 4. As the result for patients with diabetes from age 60 – 79 is listed independently to  $f_2$ , the sensitivity of  $f_1$  is still equal to 1.

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TABLE 5  
Differentially Private Data Publishing Problem Characteristics

| Differentially private data publishing |                                                                                        |
|----------------------------------------|----------------------------------------------------------------------------------------|
| The nature of input data               | transaction, histogram, graph, stream                                                  |
| The nature of output data              | query result, synthetic dataset                                                        |
| Publishing setting                     | interactive, non-interactive                                                           |
| Publishing mechanism                   | Laplace/exponential, query separation, transformation, iteration, dataset partitioning |
| Challenges                             | query number, accuracy, computational efficiency                                       |

The total noise of two queries will be  $3 * Lap(1/\epsilon)$ , which is lower than the non-interactive Laplace mechanism. The new structure is used to decompose the correlation between queries, so the sensitivity can be decreased as well. The challenge is finding a new structure.

- *Dataset partitioning*: the original dataset is divided into several parts to decrease noise. In the above example, suppose we need to answer  $f_1$  with Table 4, the noise  $Lap(1/\epsilon)$  needs to be added twice: once to the first row and again to the second row. The total noise added is  $2 * Lap(1/\epsilon)$ . However, if we partition the dataset another way, for example, by rearranging the age range to 40 – 79, the total noise will decrease to  $Lap(1/\epsilon)$ . The challenge here is designing a partition strategy for multiple queries.
- *Query separation*: query separation assumes that a query set can be separated into several groups and that some queries can be answered in the sense of reusing noise. In the above example, if  $f_2$  has been answered,  $f_1$  can be approximately answered by doubling the answer of  $f_2$  as the age range is doubled. Query separation is used to break limitations on the number of queries.
- *Iteration*: iteration is a mechanism that updates a dataset recursively to approximate the noisy answers for a set of queries. For example, we can manually define an initial dataset  $D_0$ , in which the number of patients with diabetes in Table 4 at different age ranges are equal, then perform  $f_1$  on  $D_0$ , and compare the noise result  $\hat{f}_1(D)$  with  $f_1(D_0)$ . If the distance between the two answers is smaller than a pre-defined threshold,  $f_1(D_0)$  can be published, and  $D_0$  will be used in the next round. Otherwise,  $\hat{f}_1(D)$  will be published and  $D_0$  will be updated by a particular strategy into  $D_1$ . As publishing  $f_1(D_0)$  does not consume any privacy budget, the iteration mechanism can achieve a higher utility and can answer more queries than the Laplace mechanism. The challenges are designing an update strategy and setting the related parameters, such as threshold.

Table 6 compares the various publishing mechanisms. In the following sections, we present how those mechanisms work in both interactive and non-interactive settings.

### 3.2 Interactive Publishing

Interactive settings operate on various aspects of the input data, including transactions, histograms, streams and graph datasets. In the following sections, we discuss publishing scenarios involving these types of input data.

TABLE 6  
Differentially Private Data Publishing Mechanism Comparison

| Mechanism            | Description                                                                                       | Advantage                                                                                                                                               | Challenge                                                                                                            |
|----------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Laplace              | Add Laplace noise directly to the query                                                           | Easy to implement; can answer all types of real-value queries                                                                                           | Answer a sub-linear in $n$ number of queries; introduce large volume of noise                                        |
| Transformation       | Transfers original dataset to a new structure. The sensitivity of the query set will be adjusted. | Noise can be reduced; consistency of results can be maintained                                                                                          | Not easy to find a new structure that suits for queries.                                                             |
| Dataset partitioning | Divides dataset into several parts and adds noise to each part separately.                        | Sensitivity can be decreased which results in less error in the output.                                                                                 | partition strategy is not easy when answering multiple queries                                                       |
| Iteration            | Updates datasets or query answers recursively to approximate the noisy answer.                    | Only some updates will consume privacy budget, so more queries can be answered (linear to $n$ or exponential number of $n$ ) in a fixed privacy budget. | Most iteration mechanisms are computationally inefficient; unsuitable parameters can result in inferior performance. |
| Query separation     | Only need to add noise to small numbers of queries                                                | Some queries do not consume privacy budget, so the mechanism can answer more queries with a fixed privacy budget.                                       | Separating queries is a difficult problem                                                                            |

### 3.2.1 Transaction Data Publishing

The most popular representation of  $D$  is a transaction dataset, in which every record represents an individual with  $d$  attributes.

*Query separation.* The goal of query separation is to design a separation strategy for given types of queries to decrease noise. Roth [16] presented the *median* mechanism and found that, among any set of  $m$  queries, there are  $O(\log m \log |\mathcal{X}|)$  queries that can determine the answers of all other queries. Based on this observation, all queries are separated into hard and easy queries. Hard queries can be answered directly by the Laplace mechanism, while easy queries are answered by the median values of hard query results. Therefore, easy queries do not consume any privacy budget. By separating the queries, the median mechanism can answer exponentially many more queries with acceptable accuracy; however, it is inefficient and comes with an exponential time complexity corresponding to the dataset size  $n$ .

*Iteration.* Hardt et al. [17] proposed *private multiplicative weights* (PMW), which considers datasets as a histogram with positive weight on each bin. By updating the weights, PMW constructs a histogram sequence to answer a set of queries. After the parameters have been calibrated for complexity and accuracy, this mechanism is able to answer each query with a sampling error approximately to  $O((\log m)/\sqrt{n})$ . This means that the sampling error grows logarithmically with an increase in the number of queries being answered, while the Laplace mechanism's error is linear, increasing by  $m$ . In addition, PMW can accurately answer an exponential number of queries.

Similarly, Gupta et al. [18] presented a general iteration framework, termed iterative database construct (IDC), which implements other release mechanisms by using the framework. In each round of iteration, when a significant difference between the current dataset and the original dataset is witnessed for a given query, the mechanism updates the current dataset for the next update. IDC is a more general framework that can be incorporated into various other mechanisms, including PMW and the median.

*Discussion.* The query separation and iteration mechanisms can answer more queries than that of the Laplace

mechanism, normally an exponential number of  $n$ . With an increase of the number of queries, the error bound of the Laplace mechanism increases by  $m \log m$ , whereas other mechanisms limit the increase to  $\log m$ , which is a huge improvement when searching an entire data universe. With the exception of the Laplace mechanism, most of the other mechanisms are inefficient, as they do need to traverse the whole data universe. Accordingly, the error bound grows logarithmically with  $|\mathcal{X}|$  in these mechanisms.

Given iteration mechanisms outperform others in terms of the error bound, many subsequent works, therefore, have followed the iteration mechanism approach. For example, based on IDC, Huang et al. [19] considered the distance query defined over an arbitrary metric. Iteration mechanisms are normally used in low-dimensional datasets or histograms; however, publishing high-dimensional datasets still needs further exploration.

### 3.2.2 Histogram Publishing

It is often convenient to regard transaction data in terms of their histogram representations. A histogram has  $N$  bins, and a differential privacy mechanism aims to hide the frequency of each bin. Table 4 can be considered as a histogram presentation of Table 3 with four bins. The advantage of histogram representation is that limits the sensitivity to noise [4]. For example, when the histogram serves to support the range or count queries, adding or removing a single record will affect, at most, one bin. Hence, range or count queries on the histogram have a sensitivity equal to 1, and the magnitude of added noise to each bin will be relatively small.

*Laplace.* This is a direct mechanism which adds Laplace noise to the frequency of each bin that a query covered. When a count of range query covers only small number of bins, this mechanism retains high utility for the query result; however, if the original dataset contains multiple attributes, the combination of these attributes and their related range of values will lead to a large number of bins. The answer of queries are meaningless due to the large amount of error accumulated from having an enormous number of bins.

**Dataset partitioning.** As the number of bins is derived from the partition of attribute values, one method for decreasing error is to optimize the partition strategy. For example, when the query covers a larger amount of bins, several bins can be merged into a single new bin. Laplace noise can be reduced because it is only added once to the new merged bin. When the number of bins covered by a query is relatively small, the curator can either split the large bin into smaller bins, or approximate the query result by estimating proportion of the large bin's frequency. However, splitting the large bin into smaller bins leads to more Laplace noise, while estimating the proportion of the large bin's frequency may introduce estimation error. Therefore, optimizing the partition strategy to obtain a trade-off between splitting and merging bins is a challenge that needs to be addressed.

Xu [20] provided two partition strategies by minimizing the sum of squared error (SSE) of a set of queries. Both strategies set each attribute as a single bin at an initial state and partition the attribute value to create more bins. The first strategy, *NoiseFirst*, injects Laplace noise into each bin before partitioning the attribute values. Another strategy, *StructureFirst*, uses an exponential mechanism to select optimal splitting values of attributes by adopting the SSE as the score function.

Qardaji et al. [21] partitioned the attribute values in a hierarchical way. They also focus on range queries and allocate ranges into a tree. The root of the tree is the full range of values of an attribute or several attributes. Each node in the tree is associated with the union of the ranges of its children. Several unit-length ranges are defined as leaves. On each branch of the tree, a factor controls the accuracy of the query result. These factors are further studied with a mean squared error (MSE) when answering range queries, and the results are optimized by tuning these factors.

**Histogram consistency.** Even though partitioning is a popular mechanism in histogram releases, this mechanism may bring problems with inconsistency. For example, after adding noise, the summation of two bin's value may be less than one bin. To maintain consistency in histogram publishing, Hay et al. [22] defined a *constrained inference* to adjust the publishing output. Two types of consistency constraints were explored. The first, sorted constraints, requires query results to satisfy a particular sequence. The second, hierarchical constraints, predefines the sequence of a hierarchical interval set. A constrained inference step applies a linear combination method to estimate a set of approximate answers that are close to the noisy answers, which satisfies the consistency constraints.

There are some other ways to improve the consistency of the histogram. For example, Lin et al. [23] viewed a set of sorted histograms as a Markov chain and proposed an algorithm that applies ordering constraints on the estimates. Lee et al. [24] added a post-processing step, formulated as a constrained maximum likelihood estimation problem, before publishing the histogram.

### 3.2.3 Stream Data Publishing

In real world scenarios, it is more practical to publish continuously updated data. This type of data can be simplified as a bit string  $\{0, 1\}^n$  and each 1 in the stream represents the occurrence of an event [25]. A differentially private mechanism releases one bit at every time step.

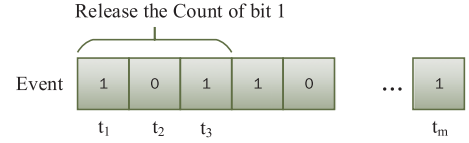


Fig. 3. Stream data.

Fig. 3 illustrates an example. Suppose there is a binary bit stream  $D \in \{0, 1\}^T$ , where  $T$  represents a time sequence  $T = \{t_k : k = 0, \dots\}$ . The bit  $\sigma(t_k) \in \{0, 1\}$  denotes whether an event has occurred at time  $t_k$ . At each time step  $t_k$ , the count of 1s is denoted as  $f(t_k)$  and the noisy output of the mechanism is  $\hat{f}(t_k)$ .

**Dataset partitioning.** Partitioning the dataset in stream data aims to recursively split the stream into sub-domains and evaluate a noisy count for each sub-domain. The problem lies in defining the sub-domain and how many times the stream can be divided.

Chan et al. [26] presented a *p-sums* mechanism which computes the partial sum of consecutive bits in a stream. Each sub-domain is defined by p-sum, which is an intermediate result estimating the count at each time interval. Laplace noise is added to a p-sum result rather than an individual count answer. This guarantees an error bound of  $O(\frac{\log^{3/2}[T]}{\epsilon})$ , which decreases the linear complexity of noise to log complexity.

Zhang et al. [27] created a quadtree for spatial datasets. They defined a threshold to determine the minimum of a sub-domain and another threshold to limit the height of a quadtree. Using these two thresholds, the amount of noise added to the quadtree can be limited to a constant.

**Iteration.** Dwork et al. [28] proposed a continual output transformation algorithm and developed an iteration release mechanism to output the continual count at each time step. The error is decreased to  $O(\frac{\log^{3/2}[T]}{\epsilon})$ .

Georgios et al. [29] proposed the notion of *w-event* privacy to achieve a balanced privacy target. They formulated the iteration mechanism using a sliding window methodology. This *w-event* privacy aims to mask any event sequence within  $w$  time steps, which is suitable for infinite stream releases.

**Discussion.** Even though continual publishing is a popular topic in the data mining community, there are still many unsolved problems in the privacy preserving area. For example, releasing a multi-dimensional dataset periodically, and dealing with other statistical queries needs further exploration.

### 3.2.4 Graph Data Publishing

With the significant growth of online social networks (OSNs), graph data has raised concern among OSN participants. An OSN dataset can be modeled as an undirected graph  $G = (V, E)$ . We use  $V$  to denote nodes and use  $E \subseteq V \times V$  to denote edges. In an OSN, nodes normally represent individuals while edges denote their relationships. Two concepts of differential privacy in graph data have been defined: *node differential privacy* and *edge differential privacy*.

**Edge Differential Privacy.** Edge Differential Privacy ensures that a query's answer does not reveal the inclusion or removal of a particular edge in the graph. Two graphs are neighboring if they differ in one edge. The first differential privacy research into graph data was conducted by Nissim [30], who showed how to evaluate the number of



TABLE 7  
Batch Query Release Example

| Query    |       |   |       |   |       |   |       |
|----------|-------|---|-------|---|-------|---|-------|
| $f_1$    | $x_1$ | + | $x_2$ | + | $x_3$ | + | $x_4$ |
| $f_2$    | $x_1$ | + | $x_2$ | + | $x_3$ |   |       |
| $f_3$    |       | + | $x_2$ | + | $x_3$ | + | $x_4$ |
| $f_4$    | $x_1$ | + | $x_2$ |   |       |   |       |
| $f_5$    |       | + | $x_2$ | + | $x_3$ |   |       |
| $f_6$    |       |   |       |   | $x_3$ | + | $x_4$ |
| $f_7$    | $x_1$ |   |       |   |       |   |       |
| $f_8$    |       |   | $x_2$ |   |       |   |       |
| $f_9$    |       |   |       |   | $x_3$ |   |       |
| $f_{10}$ |       |   |       |   |       |   | $x_4$ |

triangles in a social network with edge differential privacy. They use local sensitivity, which limits the sensitivity tightly with an upper bound, and show how to efficiently calibrate the noise for subgraph counts accordingly. The results of this technique were investigated by Karwa et al. [31] to release counts on  $k$ -triangles and  $k$ -stars. They achieved  $\epsilon$ -differential privacy for  $k$ -star counting, and  $(\epsilon, \delta)$ -differential privacy for  $k$ -triangle counting.

Zhang et al. [32] claim that if one can find an isomorphic graph with proper statistical properties that is similar to the original graph, the isomorphic graph can be used to generate accurate query answers. Given a subgraph  $G'$ , they adopted an exponential mechanism to search a number of isomorphic copies of  $G'$  to answer subgraph queries.

**Node differential privacy.** In node differential privacy, two graphs are considered to be neighboring if one of them is generated by removing one node and all the edges related to this node from the other graph. Node differential privacy is more strict than edge differential privacy. The sensitivities resulting from the change of one node are proportional to the graph size.

With the observation that many useful statistics have low sensitivities on graphs  $G_\theta$  with a small degree  $\theta$ , a common approach to achieve *node differential privacy* is to transform the graph  $G$  to a  $\theta$ -degree-bounded graph  $G_\theta$ , in which the nodes with degrees of more than  $\theta$  are deleted [33], [34].

Chen et al. [35] proposed an iteration mechanism for node differential privacy. Given a graph  $G$  and any real-valued function  $f$ , they defined a sequence of real-valued functions  $0 = f_0(G) \leq f_1(G) \leq \dots \leq f_m(G) = f(G)$  with a *recursive monotonicity* property. The recursive approach will return subgraph counting for any kind of subgraph with node differential privacy. However, constructing the sequence of functions  $f_i(G)$  is usually NP-hard, and efficiently implementing it remains an open problem.

**Discussion.** Existing methods work reasonably well with edge differential privacy or even node differential privacy for basic graph statistics. However, releasing specific statistics, such as cuts, pairwise distances between nodes, or on hyper-graphs, still remain open issues.

### 3.3 Non-Interactive Publishing

Non-interactive settings mean all queries are given to the curator at one time. The key challenge for non-interactive publishing is the sensitivity measurement. Correlation between queries will dramatically increase the sensitivity. Two possible methods are proposed to fix this problem: one is decomposing the correlation between batch queries,

TABLE 8  
 $m$  Range Queries

| query |       |   |       |   |       |   |       |
|-------|-------|---|-------|---|-------|---|-------|
| $f_1$ | $x_1$ | + | $x_2$ | + | $x_3$ | + | $x_4$ |
| $f_2$ | $x_1$ | + | $x_2$ | - | $x_3$ | - | $x_4$ |
| $f_3$ | $x_1$ | - | $x_2$ |   |       |   |       |
| $f_4$ |       |   |       |   | $x_3$ | - | $x_4$ |

which is presented in Section 3.3.1, another is publishing a synthetic dataset with the constraint of differential privacy to answer those proposed queries. Related methods are presented in the synthetic dataset publishing section.

#### 3.3.1 Batch Query Publishing

Batch query publishing refers to the most common non-interactive scenario in which a fixed set of  $m$  queries  $F = \{f_1, \dots, f_m\}$  are answered in a batch.

Using Table 4 again to show the batch query problem, suppose a curator would like to release a batch of range queries for Table 4, and Table 7 contains all possible range queries  $F = \{f_1, \dots, f_{10}\}$ . Deleting any record in  $D$  will change at most 6 query results (column containing  $x_2$  or  $x_3$  in Table 7) in  $F$ . According to the definition of sensitivity, the sensitivity of  $F$  is 6, which is much higher than the sensitivity of a single query. Therefore, most researches focus on how to decrease the sensitivity of  $F$ .

**Transformation.** One possible means of decreasing noise is to re-calibrate the sensitivity in a new data structure  $A$ .

Xiao et al. [36] proposed a wavelet transformation, called *Privelet*, which applies a wavelet transformation on the frequency dataset  $D$  to generate a wavelet coefficient  $A$ . Each entry of  $A$  is considered as a linear combination of the entries in  $D$ . Range queries can be generated with a linear combination of basic queries. Table 8 shows the basic queries. For example, the answer of  $range(x_2, x_3) = x_2 + x_3$  can be generated by  $range(x_2, x_3) = 0.5f_1 - 0.5f_3 - 0.5f_4$ , and the sensitivity will be decreased from 6 to 3. In this way, the sensitivity of the wavelet coefficient is estimated as  $1 + \log_2 n$  and the variance of noise per answer is  $O((\log_2 n)^3 / \epsilon^2)$ , which is much smaller than that in the Laplace mechanism.

Li et al. [37] proposed a *matrix* mechanism which can answer sets of linear counting queries. The set of queries, defined as a workload, is transformed into a matrix  $A$ , where each row contains the coefficients of a linear query. The essential element of the matrix mechanism is to select  $A$  to represent the set of queries. Based on the selection of  $A$ , the matrix mechanism can be extended to a variety of approaches. For example, if  $A$  is an identity matrix, this mechanism can be considered to be a normal Laplace mechanism for batch queries. If  $A$  is selected using a Haar wavelet, it can be extended as a Privelet [36]. The error bound of the matrix mechanism was analyzed in their subsequent paper [38]. They proved that when batch queries are denoted in a matrix manner, the minimum error for the result can be evaluated according to the spectral properties of this query matrix.

Similarly, Huang et al. [39] transformed the query sets into a set of orthogonal queries to reduce the correlation between queries. Yuan et al. [40] formulated searches of  $A$  as a constrained optimization problem. This approach is challenging, however, as the optimization objective is non-convex. They formulated searches of  $A$  as a low-rank matrix

factorization problem using an iteration method to approximate a suitable  $A$ . Later, they considered approximate differential privacy and transformed the optimization objective into a convex program to minimize the overall error in the results [41].

**Dataset partitioning.** Kellaris et al. [42] decomposed the dataset columns into disjoint groups and added Laplace noise to these groups' counts. The final result is generated using the new column count. Because the maximum number of original counts in a group affected by a user is limited, the sensitivity of each group is decreased, and the Laplace noise required for  $\epsilon$ -differential privacy also diminishes. The advantage of this mechanism is that it can limit the sensitivity for numerical datasets.

**Iteration.** Iteration has also been used in the batch query publishing. By recursively approximating the true answer, the noise in the output can be effectively diminished. Xiao et al. [43] aimed to decrease errors in the released output. They argued that the Laplace mechanism adds noise with a constant range to every query answer without considering the true value of the answer. Thus, queries with small answers have a much higher error than expected, defined as relative error. In practice, larger answers can tolerate more noise. In some applications, relative errors are more important than absolute errors. To decrease the relative error, Xiao et al. [43] proposed a mechanism named *iReduct*, which initially obtains rough error estimations for query answers and subsequently uses this information to iteratively refine these error evaluations.

**Discussion.** The key problem in batch query publishing is how to decrease the sensitivity between correlated queries. Currently, transformation is the most popular way to tackle this problem. Current works mainly focus on range queries and develop appropriate structures to answer linear combinations of those queries. More types of structures need to be developed to answer various types of queries. Iteration or partitioning the dataset may not be effective for correlation decomposition, but they have the potential to answer more types of queries.

### 3.3.2 Synthetic Dataset Publishing

Synthetic dataset publishing investigates publishing datasets with the public instead of sharing answers to queries. Suppose the input dataset is  $D$  with  $d$  attributes, and the curator would like to publish a synthetic  $\hat{D}$  that can be used to answer a query set  $F$ . Two different methods to achieve this goal exist. One method applies anonymization techniques to publish an anonymized dataset, which retains the same records as the original dataset. Another method takes samples from the data universe to build a synthetic dataset, which follows the original datasets distribution but may not necessarily keep the same records.

**Synthetic dataset publishing based on anonymization.** This line of works observes that if the anonymization process follows the requirement of differential privacy at each step, the published synthetic dataset will satisfy differential privacy. Based on this observation, Mohammed et al. [44] proposed the anonymized algorithm *DiffGen* to preserve privacy for data mining purposes. The anonymized procedure consists of two major steps: partitioning and perturbation. Every attribute of the original dataset is generalized to its top-most state. The partitioning step then splits these attributes into more specific groups according to the attribute taxonomy trees. It

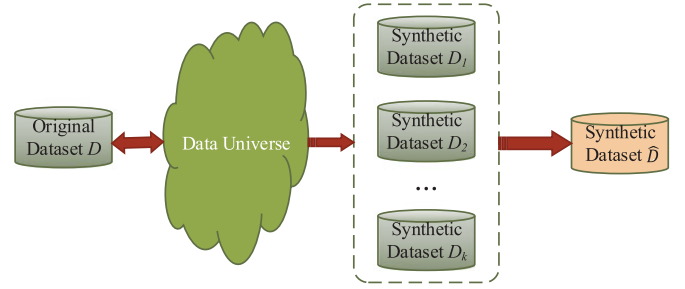


Fig. 4. Synthetic dataset publishing with learning theory.

applies an exponential mechanism to choose a candidate for the specialization. After that, random noise is added to the true count of each records group by perturbation.

**Synthetic dataset publishing based on learning theory.** Kasiviswanathan et al. [45] and Blum et al. [14] claim that if a query on a published dataset is limited to a particular concept set  $\mathcal{C}$ , the learning process can ensure the accuracy of the query answer in the constraint of differential privacy.

Kasiviswanathan et al. [45] proposed an exponential based mechanism to search a synthetic dataset from the data universe that can accurately predict  $\mathcal{C}$ . Fig. 4 shows that after creating multiple candidate datasets from the data universe, the mechanism will search the most suitable  $\hat{D}$  based on the exponential mechanism. The authors claimed that for any  $\mathcal{C}$  and any  $D \geq \{0, 1\}^d$ , if the size of the dataset satisfies

$$n \geq O\left(\frac{dVC(\mathcal{C})\log\left(\frac{1}{\alpha}\right)}{\epsilon\alpha^3} + \frac{\log\frac{1}{\beta}}{\epsilon\alpha}\right),$$

$\alpha$  accuracy can be achieved with a probability of  $1 - \beta$ .

Subsequent works made progress toward improving accuracy. Dwork et al. [46] used *boosting* to improve the lower bound of accuracy to  $O\left(\frac{\sqrt{n\log|\mathcal{X}|\log^{2/3}m}}{\epsilon}\right)$ . Hardt et al. [47] combined an exponential mechanism with a multiplicative weight iteration approach to achieve a nearly optimal accuracy guarantee.

Although these mechanisms developed tight bounds on accuracy, they suffer from inefficient computation time. A normal time cost is exponential in the size of universe  $\mathcal{X}$  and the size of concept class  $\mathcal{C}$ . Blum et al. [14] claim that if polynomial time is required, the definition of privacy has to be relaxed. This claim was confirmed by Dwork et al. [48], who proved that only after relaxing the notion of  $\epsilon$  to  $(\epsilon, \delta)$  can computational time be polynomial.

Moreover, Ullman et al. [49] showed that among all computationally efficient algorithms, the Laplace mechanism is almost optimal. In fact, no algorithm has the ability to answer more than  $O(n^2)$  queries randomly in polynomial time. This result suggests that it is difficult to design mechanisms to answer arbitrary queries efficiently.

Table 9 summarizes the key works on synthetic datasets. For simplicity, sensitivity is predefined as 1, the dependence on  $\beta$  is omitted and the run-time only considers the size of the universe.

Learning theory extends the research work on synthetic data releases, proving it is possible to maintain acceptable utility while preserving differential privacy. Nevertheless, the issue of reducing the computational complexity remains a challenge.



TABLE 9  
Synthetic Dataset Publishing with Learning Theory

| Mechanism         | Accuracy                                                                           | Efficiency  | Privacy              |
|-------------------|------------------------------------------------------------------------------------|-------------|----------------------|
| Blum et al. [14]  | $O\left(\frac{n^{2/3} \log^{1/3} m \log^{1/3} \mathcal{X}}{\epsilon^{1/3}}\right)$ | inefficient | $\epsilon$           |
| Dwork et al. [48] | $O\left(\frac{n^{1/2} \log^{1/2}  \mathcal{X}  m}{\epsilon}\right)$                | inefficient | $(\epsilon, \delta)$ |
| Dwork et al. [46] | $O\left(\frac{n^{1/2} \log^{1/2}  \mathcal{X}  \log^{3/2} m}{\epsilon}\right)$     | inefficient | $(\epsilon, \delta)$ |
| Hardt et al. [47] | $O\left(\frac{\log  \mathcal{X}  \log m}{\epsilon n}\right)^{1/3}$                 | inefficient | $\epsilon$           |

*Synthetic dataset publishing for high dimensional datasets.* Neither anonymized dataset publishing nor learning theory-based publishing can effectively handle high-dimensional datasets. In anonymized methods, when the input dataset contains many attributes, existing anonymized methods will inject a prohibitive amount of noise, which leads to inferior utility. In learning theory-based methods, computational complexity is exponential to the dimension of the dataset, making the publication infeasible for high-dimensional datasets. One promising way to address high dimensionality is to disassemble the dataset into a group of lower dimensional marginal datasets, and then apply methods to infer the joint data distribution from these marginal datasets.

Zhang et al. [50] followed the above rationale and used a Bayesian network to deal with high dimensionality. They assumed some correlations between attributes exist and, if these correlations can be modeled, the model can be used to generate a set of marginal datasets to simulate the distribution of the original dataset. The disadvantage of this solution is that it consumes too much of the privacy budget during network construction and, hence, makes the approximation of the distribution inaccurate.

Chen et al. [51] addressed this disadvantage by proposing a clustering method. They disclose the pairwise correlation of all the attributes and generate a dependency graph. A junction tree algorithm is applied to the graph to identify a set of attribute clusters, which are used to generate noisy marginals. As a final step, an inference model is used to create a synthetic dataset. As opposed to [50], they have limited access to the dataset, saving the privacy budget to obtain a better result.

### 3.4 Summary of Differentially Private Data Publishing

#### 3.4.1 Summary of the Interactive Setting

The interactive setting has attracted attention due to advances in statistical databases. In interactive settings, the privacy mechanism receives a user's query and replies with a noisy answer to preserve privacy. Traditional Laplace mechanisms can only answer  $O(n)$  queries, which is insufficient in many scenarios. Researchers have to provide different mechanisms to fix this essential weakness.

The proposed interactive publishing mechanisms improve performance in terms of query type, the maximum number of queries, accuracy, and computational efficiency. Upon analysis, we conclude that these measurements in interactive releases are associated with one another. For example, given a fixed privacy budget, a higher accuracy usually results in a smaller number of queries. On the other hand, with a fixed accuracy, a larger number of queries normally leads to computationally inefficient mechanisms.

Therefore, the goal of data publishing mechanism design is to achieve a better result that can balance the above mentioned measurements. The choice of mechanism depends on the requirement of the application.

#### 3.4.2 Summary of the Non-Interactive Setting

High sensitivity presents a big challenge in non-interactive settings. Batch query publishing methods can only publish limited types of queries. Publishing a synthetic dataset seems appealing because, in some scenarios, people require details of the attributes to determine further analysis methods. The research on synthetic data publishing, however, is still in its early stages and there are many open problems in this area. The essential problem is efficiency. Given most publishing mechanisms need to sample datasets from the entire data universe, it is hard to search for a suitable dataset in polynomial time.

Another problem is that synthetic dataset publishing can only publish datasets for particular purposes. For example, an anonymization dataset focuses on a decision tree algorithm, the published dataset obtains an acceptable result for decision tree tasks, yet the proposed method does not guarantee the performance for other types of tasks. Learning-based methods have the same disadvantage, or worse, as they only guarantee learning performance for a particular class. Publishing a dataset for multiple purposes needs further investigation.

The third problem is dealing with high-dimensional datasets. Even though [50] and [51] have undertaken some initial work, they both consume too much of the privacy budget when building the distribution model, making the results less accurate than that of lower-dimensional datasets.

## 4 DIFFERENTIALLY PRIVATE DATA ANALYSIS

The essential task of differentially private data analysis is extending the current non-private algorithms to differentially private algorithms. This extension can be realized by several frameworks, roughly categorized into Laplace/exponential frameworks and private learning frameworks. The Laplace/exponential framework incorporates Laplace or exponential mechanisms into non-private analysis algorithms directly. For example, adding Laplace noise to the count steps in the algorithm or by employing perform exponential mechanisms when making selections.

Private learning frameworks consider the data analysis as learning problems in terms of optimization. The learning problems are solved by defining a series of objective functions. Compared with the Laplace/exponential framework, a private learning framework has a clear target, and the results produced by this framework are easier to compare in terms of risk bound or sample complexity. But private learning frameworks can only deal with limited learning algorithms, while nearly all types of analysis algorithms can be implemented in a Laplace/exponential framework.

Table 10 shows the characteristics of the differentially private data release problems. Researchers are concerned with the accuracy and computational efficiency of these two frameworks. As different papers use diverse terms to describe the output, the terms "model" and "algorithm" are interchangeable in this section.

### 4.1 Laplace/Exponential Framework

The most common extension method is to incorporate Laplace or exponential mechanisms into non-private analysis algorithms. These algorithms are usually associated with

TABLE 10  
Differentially Private Data Analysis Problem Characteristics

| Differentially Private Data Analysis |                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------|
| The nature of input data             | Transaction                                                                                                      |
| The nature of output data            | Analysis models/algorithms                                                                                       |
| Analysis framework                   | Laplace/exponential framework, private learning framework                                                        |
| Analysis mechanism                   | Laplace/exponential mechanism for Laplace/exponential framework, learning process for private learning framework |
| Challenges                           | Accuracy, computational efficiency                                                                               |

specific machine learning or data mining tasks, which are separated into the supervised learning, unsupervised learning, and frequent pattern mining categories.

#### 4.1.1 Supervised Learning

Supervised learning refers to the prediction methods that extract models describing data classes via a set of labeled training records [52]. As one of the most popular supervised learning algorithms, *decision tree* learning, has been extensively studied in Laplace/exponential Frameworks.

Decision trees are iterative processes that recursively partition the training sample to build a tree with each label representing a leaf. Assuming there is an input dataset  $D$  with  $d$  categorical attributes  $\{a_1, \dots, a_d\}$ , a decision tree is constructed from the root that holds all the training records then the algorithm chooses the attribute  $a_i$  that maximizes the information gain to partition the records into child nodes. The procedure is performed recursively on each subset of the training records until a stop criteria is met.

The first differentially private decision tree algorithm was developed on the SuLQ platform [53]. Noise is added to the information gain, and an attribute  $a_i$  with noisy information gain that is less than a specified threshold is chosen to partition a node. However, as information gain is evaluated separately for each attribute in each iteration, the privacy budget is consumed several times in each iteration, which results in a large volume of noise. In addition, SuLQ fails to deal with continuous attributes. If those attributes are simply discretized into intervals, the basic concept of differential privacy is violated, because the split values in continuous attributes would reveal information about the records.

To overcome the SuLQ platform's disadvantages, Friedman et al. [54] improved the algorithm in two ways. First, they implemented an exponential mechanism in the attribute selection step. The score function is defined by the information gain or the gain ratio. The attributes with a top score have a higher probability of being selected. In this way, less of the privacy budget is consumed than by SuLQ. Second, the proposed method can deal with continuous attributes. An exponential mechanism is employed to select every possible splitting value and the continuous attributes domain is divided into these intervals. Compared to SuLQ, they obtain better performance.

Jagannatham et al. [55] provided an algorithm for building random private decision trees, which randomly select attributes to create nodes. The algorithm first creates a tree in which all the leaves are on the same level and then builds a leaf count vector. Once the independent Laplace noise is

added to the count vector, a differentially private random decision tree can be generated from the noisy leaf vector. The algorithm iteratively produces multiple random decision trees and uses the ensemble method to combine these trees. As the attribute is randomly selected, this step saves the privacy budget; however, as each tree's magnitude is scaled up with the number of trees in the final ensemble step, the utility of the ensemble remains a problem.

Rana et al. [56] proposed a practical approach to ensemble decision trees in a random forest. They do not strictly follow the notion of differential privacy, which keeps the neighboring data distribution approximately invariant. Instead, they only keep the statistic features invariant. A privacy attack model is defined to prove the reliability of the proposed decision tree. As less budget has been consumed in the ensemble process, this relaxation of differential privacy can lead to higher utility compared to other algorithms.

*Discussion.* Table 11 shows a summary of differential private supervised learning methods. Differentially private decision tree algorithms were the earliest algorithms to be investigated in the differential privacy community. The advantage of this series of methods is that they are concise and easy to implement. However, because tree-based algorithms need to select split attributes multiple times, the privacy budget is quickly consumed, which incurs a huge utility loss. This drawback stems from decision tree building and is not easy to deal with. Nowadays, one of the most popular ways to design differentially private supervised learning algorithms is to apply a private learning framework, which is discussed in Section 4.2.

#### 4.1.2 Unsupervised Learning

As a typical unsupervised learning method, *clustering* algorithms group unlabeled records into clusters to ensure that all records in the same cluster are similar to each other. Assume the input is a set of points,  $r_1, \dots, r_n$ , the clustering output is  $k$  centers  $\{v_1, \dots, v_k\}$  and assigned points.

##### Algorithm 1. Basic k-Means Algorithm

**Require:** points  $r_1, \dots, r_n$ .

**Ensure:**  $k$  centers  $\{v_1, \dots, v_k\}$  with arranged points

**repeat**

1. Create  $k$  clusters by assigning each point to its closest center;

2. Re-calculate the center of each cluster;

**until** centers do not change

A basic  $k$ -means algorithm is formulated by Algorithm 1. The aim of differential privacy clustering is to add uncertainty into the center  $v$  and the number of records in each center. To achieve this goal, noise is added in Step 1 in Algorithm 1. In fact, adding noise to cluster centers is impractical because the sensitivity of the cluster center will be quite large as deleting one point will totally change the center. Therefore, the challenge for clustering is to evaluate and minimize the sensitivity of the cluster centers.

Nissim et al. [30] used local sensitivity with  $k$ -means clustering to circumvent the large sensitivity problem, by relying on the following intuition. In a well-clustered scenario, a point with noise should have approximately the same center as its previous center. In addition, moving a few "well-clustered" records would not ultimately change the centers. As such, they define a local sensitivity to measure the record-

TABLE 11  
Comparison of Supervised Learning Methods

| Difficulty                                       | Key methods                                   | Typical papers | Advantages                                                 | Disadvantages                                                               |
|--------------------------------------------------|-----------------------------------------------|----------------|------------------------------------------------------------|-----------------------------------------------------------------------------|
| Privacy budget has to be consumed multiple times | Add noise to information gain                 | [53]           | Easy to implement                                          | Noise will be high due to privacy budget arrangement in attribute selection |
|                                                  | Use exponential mechanism to select attribute | [54]           | Save part of privacy budget in the attribute selection     | Still has high noise                                                        |
|                                                  | Select attribute randomly                     | [56], [55]     | Does not consume privacy budget during attribute selection | Privacy budget will be largely consumed in the ensemble process             |

based sensitivity of the cluster center, which is much lower than the traditional global sensitivity. Since the value of local sensitivity is difficult to measure, they provide a sample aggregate method to approximate local sensitivity.

Based on local sensitivity, Wang et al. [57] implemented subspace clustering algorithms. They introduce Laplace mechanism into an agnostic  $k$ -means clustering, and an exponential mechanism into Gibbs sampling subspace clustering algorithm. Their subsequent paper, [58] adopted a Johnson-Lindenstrauss transform to guarantee differential privacy in a subspace clustering algorithm. The Johnson-Lindenstrauss transform can reduce the dimensions of the dataset, making the clustering problem practical, as well as preserving the distance between each record.

*Discussion.* In general, even some differentially private platforms such as SuLQ [53], PINQ [12], Privgene [59], Gupta [60] can automatically implement clustering algorithms. They all assume that the sensitivity of the cluster centers has been predefined. Even though local sensitivity can partially solve the problem, further reducing sensitivity still remains a challenge. We use Table 12 to illustrate the advantages and disadvantages of local sensitivity and Johnson-Lindenstrauss transform.

#### 4.1.3 Frequent Itemset Mining

Frequent itemset mining aims to discover itemsets that frequently appear in a dataset  $D$ . Suppose  $I$  is the set of items in  $D$ , and an *itemset* refers to a subset of  $I$ . Let each record  $r_i \in D$  denote as a transaction that contains a set of items from  $I$ . Given an itemset  $I_j$ , if transaction  $r_i$  contains  $I_j$ , we say  $r_i$  supports  $I_j$ , and the proportion of supporting transactions in the dataset is defined as the *support* of  $I_j$ . An itemset with a frequency larger than the predefined support threshold is called a frequent itemset or a frequent pattern.

Let  $U$  represent all frequent itemsets, where the *topk* most frequent itemsets in  $U$  should be released under differential privacy guarantee. Laplace noise is normally added to the frequency; however, the main challenge is that the total number of itemsets is exponential to the number of items: If  $I$  contains

$n$  items, the number of all possible itemsets is  $|U| = \sum_{i=1}^n \binom{n}{i}$ . Decreasing the number of candidate itemsets is a major research issue in differentially private frequent itemset mining.

Bhaskar et al. [61] solved the problem by providing a truncated frequency to reduce the number of candidate itemsets. They proposed an algorithm that uses the exponential mechanism to choose the top- $k$  itemsets. The score function of each candidate is the frequency defined as  $p(\overline{U}) = \max(p(U), p_k - \gamma)$ , where  $p_k$  is the frequency of the  $k$ th most frequent itemsets and  $\gamma \in [0, 1]$  is an accuracy parameter. Every itemset with a frequency greater than  $p_k - \gamma$  is computed as its normal frequency  $p(U)$  while the frequency of the rest is truncated to  $p_k - \gamma$ . All the itemsets with frequencies of  $p_k - \gamma$  are grouped into one set, and the algorithm uniformly selects itemsets from this set. In this way, the computational cost is significantly reduced.

The advantage of truncating frequency is that it can significantly decrease the size of candidate itemsets. However, it is only applicable when  $k$  is small. Another weakness is that the length of top- $k$  itemsets needs to be predefined, which affects the flexibility of frequent itemset mining.

To address these weaknesses, Li et al. [62] constructed a basis set  $\mathcal{B} = \{B_1, B_2, \dots\}$  in which any itemset with a frequency higher than a threshold is a subset of the basis set  $B_i$ . The algorithm generated candidate itemsets based on  $\mathcal{B}$  and the algorithm could release itemsets with arbitrary length. However, generating basis sets  $\mathcal{B}$  is not easy; furthermore, when the length of the itemset and the number of basis sets are large, the cardinality of the candidate itemsets is still too big to handle. Efficiently decreasing the number of the candidate itemsets is still a challenge.

Zeng et al. [63] proposed an algorithm that randomly truncates transactions in a dataset according to a predefined maximal cardinality. The algorithm iteratively generates candidate itemsets and perturbs the support of those candidate itemsets.

Lee et al. [64] proposed an FP-tree based frequent itemset mining algorithm. The algorithm first identifies all frequent itemsets without knowing their exact supports, but only

TABLE 12  
Comparison of Unsupervised Learning Methods

| Difficulty                             | Key methods                     | Typical papers | Advantages                                                              | Disadvantages                                                                  |
|----------------------------------------|---------------------------------|----------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| High sensitivity of clustering centers | Use local sensitivity           | [30], [57]     | Decreases the level of the sensitivity                                  | Local sensitivity may not be easy to estimate                                  |
|                                        | Johnson-Lindenstrauss transform | [58]           | Guarantees differential privacy while retaining distance between points | Johnson-Lindenstrauss transform is only valid for norm 2 distance measurement. |



TABLE 13  
Comparison of Frequent Itemset Mining Methods

| Difficulty               | Key methods                         | Typical papers         | Advantages                                  | Disadvantages                                                                        |
|--------------------------|-------------------------------------|------------------------|---------------------------------------------|--------------------------------------------------------------------------------------|
| Large candidate itemsets | Merge candidates into groups        | [61], [62], [63], [64] | Easy to implement                           | Merge strategy has high impact on the output. Some important patterns may be missed. |
|                          | Binary search in the candidate set. | [66]                   | The frequent itemsets can be quite accurate | The search time can be decreased, but still inefficient.                             |
|                          | Sampling from candidate set         | [65]                   | Highly efficient                            | Some important patterns may be missed.                                               |

that their supports are above a predefined threshold. The proposed algorithm injects noise into the data structure at the intermediate (FP-tree) step. The final output can be further refined through an optional post-processing step. The advantage of the algorithm is that information disclosure affecting differential privacy occurs only for count queries above the predefined threshold; negative answers do not count against the privacy budget.

Shen et al. [65] applied a Markov chain Monte Carlo (MCMC) sampling method to deal with the challenge of large candidate itemsets. They claim that an MCMC random walk method can bypass the problem, and the exponential mechanism can then be applied to select frequent itemsets. They use this method to mine frequent graph patterns.

Xu et al. [66] applied a binary estimation method to identify all possible frequent itemsets and then use an exponential mechanism to privately select frequent itemsets. The number of candidates is eventually a logarithm of the original number using a binary search.

*Discussion.* Frequent itemset mining is a typical data mining task, which suffers from searching large candidate sets. Differential privacy makes the problem worse. [61], [63], [64] and [62] tried to merge some candidates into groups using different methods. [65] adopted a sampling method to search the candidate itemsets. [66] applied a binary search method. All of those methods decreased the searching space from exponential to polynomial, which is a big achievement. However, the noise still remains high, which needs to be decreased further. Table 13 summarizes those key methods.

#### 4.1.4 Discussion on the Laplace/Exponential Framework

The Laplace/exponential frameworks can introduce Laplace and exponential mechanisms freely to various types of algorithms. However, the utility of the analysis results is a challenge for this framework. When adding noise to algorithm steps, it is unclear how large the utility loss will be and the analysis result is not easy to compare.

One possible way to tackle the difficulty is solving the data analysis problem through the view of optimization and taking advantage of some existing theories, such as the learning theory [45], so that the utility loss can be estimated and compared. Based on this intuition, researchers proposed the private learning framework.

## 4.2 Private Learning Framework

Another line of differential privacy research investigates learning problems from the perspective of machine learning theory [67]. A learning problem is shown in Fig. 5a. Suppose

$D = \{r_1, \dots, r_n\}$  is a set of samples drawn from a universe  $\mathcal{X}$ . Dots and triangles in the figure denote two labels  $y \in \{0, 1\}$ . The function that separates one label from the other is defined as concept  $c$ . Suppose there are groups of functions (hypotheses)  $H$  and  $h_j$ , the goal of the learning process  $L$  is to find a hypothesis  $h$  that satisfies with  $c$  on almost all of universe  $\mathcal{X}$ . Fig. 5b illustrates a typical learning process. By using the input training samples, the learner selects the most suitable  $h \in H$  as the output model  $w$ .

The purpose of a private learning framework is to design a private learner that outputs an approximately accurate model and preserves the differential privacy of training samples. Private learning frameworks are concerned with the following issues:

- How to choose an optimal model in terms of differential privacy?
- How many samples are needed to achieve a bounded accuracy?

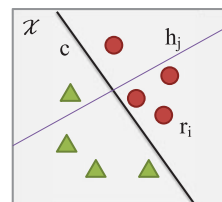
The first question can be answered by combining an Empirical Risk Minimization (ERM) technique with differential privacy. The utility is measured by a risk bound. The second question can be tackled by analyzing the sample complexity of a private learner via PAC learning theory. Both risk bound and sample complexity can be considered as implementations of Eq. (7).

### 4.2.1 Private Learning in ERM

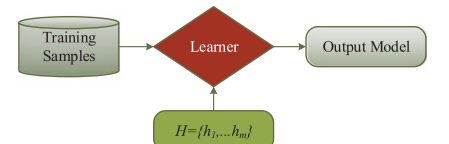
*The foundation of ERM.* ERM is used to select an optimal model from a set of hypotheses by minimizing the expected loss over the collected samples [68]. Suppose  $h \in H$  is a hypothesis and  $w$  is the output model. We define a loss function  $\ell(h(w, r), y)$  to estimate the expected risk of the hypothesis. The goal of ERM is to identify a  $w$  that minimizes the empirical risk  $R_n(w)$  on training sample  $D$ . Eq. (8) shows the empirical risk minimization

$$R_n(w) = \min_w \frac{1}{n} \sum_{i=1}^n \ell(h(w, r_i), y_i) + \lambda \mu(w), \quad (8)$$

where the regularizer  $\lambda \mu(w)$  prevents over-fitting.



(a) Learning Problem



(b) Learning Process

Fig. 5. Learning problem and process description.

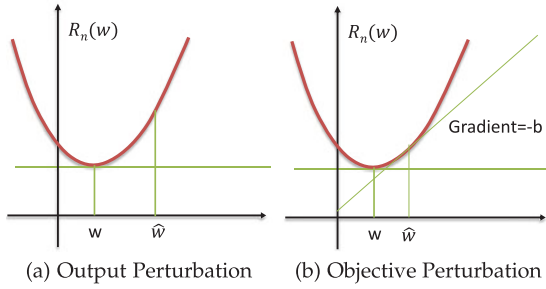


Fig. 6. Output and objective perturbation.

By choosing different loss functions, ERM can be implemented for certain learning tasks, such as linear regression [69], in which the loss function is defined as maximum likelihood estimation (MLE); logistic regression [68], in which the loss function is defined as logistic loss; and the kernel method [70], [71], in which the loss function is set to hinge loss. To make the ERM solution tractable, people assumed that 1) the loss function and regularizer are convex and 2) the loss function is  $L$ -Lipschitz. Under these two assumptions, Eq. (8) can be considered to be a  $d$  dimensional convex optimization problem.

The utility of private ERM is measured by the difference between the real risk  $R(\mathbf{w})$  and the private  $R(\hat{\mathbf{w}})$ , defined as a *risk bound*. A lower risk bound leads to a higher utility.

*Perturbation methods.* When incorporating differential privacy into the current learning process, current works apply two methods via an ERM technique: *output perturbation* and *objective operation*. The output perturbation inserts noise into the output  $\mathbf{w}$ ; while the objective operation adds noise to the objective function prior to learning.

Chaudhuri et al. [68] proposed output perturbation solutions on *logistic regression*. Fig. 6a shows that the  $\hat{\mathbf{w}}$  is obtained by moving the original  $\mathbf{w}$  on the horizontal axis. The sensitivity is  $\frac{2}{n\lambda}$ , which is associated with the regularizer parameter  $\lambda$ . With analysis, Chaudhuri et al. argued that the learning performance of this private algorithm degrades with a decreasing of  $\lambda$ . Their subsequent paper, [69], trains classifiers on disjoint subsets of the data with different parameters and uses an exponential mechanism to privately choose parameters.

The objective perturbation adds noise to the loss function  $\ell(D)$  [68], [69], which is illustrated in Fig. 6b and shows that the calibration of noise is associated with the gradient of  $R_n(\mathbf{w})$ . Chaudhuri et al. conclude that objective perturbation outperforms output perturbation when the loss function is convex and doubly differentiable. This is because regularization already changes the objective to protect against over-fitting, and changing the objective will not significantly impact performance. This claim is confirmed in terms of a risk bound in the next section.

*Risk bound in different learning algorithms.* Many machine learning algorithms can be implemented in privacy learning frameworks, including linear regression, online learning, deep learning, etc. With the exception of the learning process and the privacy budget, the risk bound is related to the dimension and the size of training samples. Bassily et al. [72] showed that the risk bound depends on  $O(\sqrt{d}/n)$  under  $(\epsilon, \delta)$ -differential privacy and  $O(d/n)$  under  $(\epsilon)$ -differential privacy. These results show that the larger the dimension and the size, the lower utility that ERM can achieve. Table 14 lists several typical learning algorithms and their risk bounds.

Nearly all types of learning problems can be analyzed in terms of their risk bound. For example, most of the existing papers solve regression problems that have a single optimal objective. Ullman [73] considered the regression problem as a multiple objective optimization when the datasets are examined with different aims. Multiple convex minimization is considered as a set of queries that can be answered by a prevalent multiplicative weights method. Ullman implemented several single objective regression results in multiple objectives platform and their risk bounds are consistent with those original papers.

Kasiviswanathan et al. [74] considered private incremental regression in terms of streaming data. They combined continual release [28] with ERM technology to analyze the risk bound of several algorithms. Taking linear regression as an example, they continuously update a noisy version of the gradient function to minimize the MLE loss function. The risk bound depends on the  $\sqrt{d}$  and the length of the stream:  $O(\min\{\sqrt{d}, T\})$ , where this bound is quite close to the normal linear regression private learner when the stream length  $T$  is large.

In addition, some papers consider private learning in a higher dimension dataset. Kifer et al. [75] gave the first results on private sparse regression with high dimensions. The authors designed the algorithm based on sub-sampling stability for support recovery using a LASSO estimator. Their following work [76] extended and improved the results with an algorithm based on a sample efficiency test of stability. Jain et al. [77] proposed an entropy regularized ERM using a sampling technique. This algorithm provides a risk bound that has a logarithmic dependence on  $d$ . Kasiviswanathan et al. [78] considered random projections in ERM frameworks. They provided a new private compress learning method with the risk bound related to the Gaussian width of the parameter space  $C$  in the random projection.

One of the most promising directions is the deep learning. Recent research focus has been devoted to the design of deep learning mechanisms. Abadi et al. [79] applied objective perturbation in a deep learning algorithm by defining the loss function as the penalty for mismatching training data. As the loss function is non-convex, they adapted a mini-batch stochastic gradient descent algorithm to solve the problem. The noise is added to every step of the stochastic gradient descent. In another pilot work [80], Shokri et al. designed a distributed deep learning model training system that enables multiple parties to jointly learn an accurate neural network. They implemented private stochastic gradient descent algorithms to achieve  $(\epsilon, \delta)$ -differential privacy within multiple parties. In the work of Pham et al. [81], the authors perturbed the objective functions of the traditional deep auto-encoder and designed the deep private auto-encoder algorithm by incorporating a Laplace mechanism.

*Discussion.* Private learning applies ERM to estimate the hypothesis set to select an optimal output  $\mathbf{w}$ . It uses the output perturbation and the objective perturbation to ensure that the output satisfies differential privacy. A series of works have proven that this private learning is tractable for certain learning tasks, such as linear regression, SVM, and deep learning, etc.

Risk bounds are highly associated with the dimension of the dataset. Current research has decreased dependence on dimension to  $O(d)$ . Under certain assumptions, the dimension dependence could be further relaxed.

TABLE 14  
Private Learning Risk Bound

| Learning algorithm | References                  | Perturbation method           | Loss function                                                      | Risk bound                            |
|--------------------|-----------------------------|-------------------------------|--------------------------------------------------------------------|---------------------------------------|
| Regression         | Chaudhuri et al. [69]       | output/objective perturbation | regularized MLE                                                    | depends on $d \log d /$ depend on $d$ |
| SVM                | Jain et al. [70]            | output/objective perturbation | arbitrary kernel, for example, polynomial kernels                  | depends on $d^{1/3} / n^{2/3}$        |
|                    | Benjamin et al. [82]        | output perturbation           | hinge-Loss function with invariant kernel, such as Gaussian kernel | depends on $\sqrt{d}$                 |
| Online learning    | Kasiviswanathan et al. [74] | objective perturbation        | regularized MLE                                                    | depends on $\sqrt{d}$ and time $T$    |
| Deep Learning      | Abadi et al. [79], [81]     | objective perturbation        | average of unmatched sample                                        | N/A                                   |

#### 4.2.2 Sample Complexity in Private Learning

The second problem: *how many samples are needed in bounded accuracy?* is associated with sample complexity analysis. Sample complexity interprets the distance utility measurement in another way to show how many samples are needed to at least achieve a particular accuracy  $\alpha$ . Probably approximately correct (PAC) learning [83] helps to measure the sample complexity of learning algorithms. Based on this theory, Blum et al. [14] and Kasiviswanathan et al. [45] proved that every finite concept class can be learned privately using a generic construction with a sample complexity of  $O(VC(\mathcal{C}) \log |\mathcal{X}|)$  (we omit the other parameters). This is a higher sample complexity than a non-private learner who only needs a constant number of samples. Improving sample complexity is an essential issue, and its study can be categorized into the following three groups: relaxing the privacy requirement, relaxing the hypothesis, and semi-supervised learning.

*Relaxing privacy requirement.* Relaxing the privacy requirement is an effective way to close the gap of sample complexity. Beimel et al. [10] showed that when relaxing  $\epsilon$ -differential privacy to  $(\epsilon, \delta)$ -differential privacy, sample complexity can be significantly decreased. Their follow-up paper decreases sample complexity to  $O(\log(\sqrt{(d \cdot 1/\delta)}))$  [84].

Another way to relax privacy requirements is to preserve the privacy of sample labels, rather than all sample attributes. Chaudhuri et al. [85] assumed that with the exception of labels, the attributes of the samples are insensitive. Label privacy aims to decrease the dimension of sensitivity information. This may provide enough protection in scenarios where the content of the underlying samples is publicly known except for their labels. In other scenarios, however, these attributes may be highly sensitive, and Chaudhuri et al.'s relaxation may not be applicable in these cases.

*Relaxing hypothesis.* This gap can also be closed by relaxing the requirement of the hypothesis. If the output hypothesis is selected from the learning concept  $H \subseteq \mathcal{C}$ , the learning process is defined as a proper learning. Otherwise, it is called an improper learning. For proper learning, the sample complexity is approximately  $\Omega(d)$ . If choosing improper learning, the sample complexity can be further decreased.

Beimel et al. [86] confirmed that when selecting a hypothesis that is not in  $\mathcal{C}$ , the sample complexity can be decreased to the constant. Their subsequent paper [87] proposed a probabilistic representation of  $\mathcal{C}$  to improve the sample complexity.

They considered a list of hypothesis collections  $\{H_1, \dots, H_m\}$  rather than just one collection of  $H$  to represent  $\mathcal{C}$ . The authors assumed that, when sampling  $H_i$  from the hypothesis list, there will be  $h \in H$  close to  $c$  in high probability. The sample complexity can be reduced to  $O(\max(\ln |H_i|))$ .

This improvement in sample complexity comes at the cost of an increased workload on evaluation, however. The learner will have to exponentially evaluate many points that are far from the concept set  $\mathcal{C}$ . In general, for a private learner, if  $H = \mathcal{C}$ , the sample complexity is  $O(d)$  and the time for evaluation is constant. If  $H \neq \mathcal{C}$ , there is constant sample complexity but  $O(\exp(d))$  time for evaluation.

*Semi-supervised learning.* Semi-supervised learning is a useful method for reducing the complexity of labeled samples. Beimel et al. [88] proposed a private learner by introducing semi-supervised learning to active learnings. The method starts with an unlabeled dataset to create a synthetic dataset for a class  $\mathcal{C}$ . This synthetic dataset is then used to choose a subset of the hypotheses with a size of  $2^{O(VC(\mathcal{C}))}$ . In the last step the authors apply  $O(VC(\mathcal{C}))$  labeled examples to choose the target synthetic dataset according to the hypotheses set.

In this process, the sample complexity of the labeled samples is  $O(\frac{VC(\mathcal{C})}{\alpha^3 \epsilon})$  while for the unlabeled samples it is  $O(\frac{d \cdot VC(\mathcal{C})}{\alpha^3 \epsilon})$ . Comparing the sample complexity of labeled and unlabeled samples, this private learner uses a constant number of labeled samples and  $O(d)$  unlabeled samples.

*Discussion.* Table 15 compares the different methods in terms of their sample complexity. To make the results clear, we use a *VC dimension* representation and omit the other parameters such as  $\alpha$ ,  $\beta$  and  $\epsilon$ .

#### 4.3 Summary of Differentially Private Data Analysis

As the most prevalent framework in differentially private data analysis, the Laplace/exponential framework has been widely used. The most prominent advantages are its flexibility and simplicity; it can freely introduce Laplace and exponential mechanisms into various types of algorithms. For the non-experts on privacy, it proposed a possible way to ensure that the results satisfy the requirement of differential privacy. However, the essential challenge for this framework is accuracy of the results, especially for those algorithms whose operations have high sensitivities. They lead to a large volume of noise in the analysis results. The Laplace/exponential framework is widely



TABLE 15  
Comparison of the Sample Complexity of Different Methods

| Method                   | References                                       | Description                                                             | Sample complexity                                                        | Privacy level        |
|--------------------------|--------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------|
| Original                 | Kasiviswanathan et al. [45],<br>Blum et al. [14] | Uses an exponential mechanism to search $h$                             | $O(VC(\mathcal{C})\log \mathcal{X} )$                                    | $\epsilon$           |
| Relaxing privacy level   | Beimel et al. [10]                               | from $\epsilon$ to $\epsilon, \delta$                                   | $O(\log(1/\delta))$                                                      | $(\epsilon, \delta)$ |
|                          | Steinke et al. [84]                              | from $\epsilon$ to $\epsilon, \delta$                                   | $O(\log(\sqrt{d \cdot 1/\delta}))$                                       | $(\epsilon, \delta)$ |
|                          | Chaudhuri et al. [85]                            | Only preserve privacy for labels                                        | $\Omega(d')$ ( $d'$ was the adjusted dimension which is lower than $d$ ) | $(\epsilon)$         |
| Relaxing hypothesis      | Beimel et al. [87] [86]                          | $H \neq \mathcal{C}$ and set a group of $H$ to privately select a $h$ . | $O(\max(\ln H_i ))$                                                      | $\epsilon$           |
| Semi-supervised Learning | Beimel [88]                                      | use labeled data to search $h$                                          | $O(dVC(\mathcal{C}))$ (labeled) $O(VC(\mathcal{C}))$ (unlabeled)         | $\epsilon$           |

used in current applications but there is still room for further improvement.

Private learning frameworks combine differential privacy with diverse learning algorithms to preserve the privacy of the learning samples. The foundation theories are ERM and PAC learning. ERM helps to select the optimal learning model by transferring the learning process into a convex minimization problem. Differential privacy either adds noise to the output models or to the convex objective functions. PAC learning estimates the relationship between the number of learning samples and the models accuracy. The more samples an algorithm can obtain, the more accurate the result will be. As privacy learning results in higher sample complexity, researchers are currently trying to narrow the sample complexity gap between private and non-private learning processes so that private learning is feasible with an acceptable number of samples.

Private learning frameworks, however, have some constraints. ERM requires that the objective function should be convex and L-Lipschitz. PAC learning can only be applied when the algorithm is PAC learnable. These constraints hinder the practical development of privacy learning frameworks and make them an actively developing theoretical proposition, yet still impractical for real applications.

## 5 APPLICATIONS OF DIFFERENTIAL PRIVACY

### 5.1 Differential Privacy in Location-Based Services

Advances in sensor-enabled devices, such as mobile phones and wearable gadgets, allow location information to be available on social media. This means that the places that people visit are able to disclose extremely sensitive information about their behaviors, home and work locations, preferences, and habits. In addition, the continual release of ones location can be used as a trajectory. Location and trajectory privacy are emerging issues that need to be tackled [89].

Location based services involve several privacy concerns. For example, a location based server would like to hide the number of people in a particular region, and this is a typical range query that differential privacy can deal with. Cormode [90] applied spatial decomposition methods, which is a type of dataset partitioning mechanism, to decrease noise. They instantiated a hierarchical tree structure to decompose a geometric space into smaller areas with data points partitioned among the leaves. Noise is added to the count for each node. Similarly, Zhang et al. [91] applied spatial decomposition methods in the problem of private location

recommendations, which is the extension of range queries. Quadtree is used to partition the region and noise is added to protect users' historical trajectories.

Some location based applications need to hide the exact locations of individuals. Chatzikokolakis et al. [92] proposed a new notion, *geo-indistinguishability*, which protects an individual's exact location, while disclosing enough location information to obtain the desired service. Its main idea relates to a differential privacy level of the radius that the individual has visited: for any radius  $\vartheta > 0$ , an individual will have  $(\epsilon, \vartheta)$ -privacy.

Chen et al. [93] presented a data-dependent solution for sanitizing large-scale trajectory data. They developed a constrained inference technique to increase the resulting utility. He et al. [94] presented a system to synthesize mobility data based on raw GPS trajectories. They discretize raw trajectories using hierarchical reference systems to capture individual movements at differing speeds. They then propose an adaptive mechanism to select a small set of reference systems to construct prefix tree counts. Lastly, a direction-weighted sampling is applied to improve utility.

Even though previous works provide various solutions to location privacy, there are still some open questions. For example, most applications require the system to release a synthetic location dataset rather than query answers, and this is a tough problem in differential privacy. Moreover, privacy issues in spatial crowdsourcing is another emerging research area [95].

### 5.2 Differentially Private Recommender Systems

Recommender systems are one of the most popular applications in e-commerce and online social networks. They are capable of recommending products users will probably like. Collaborative filtering (CF) is one of the most popular recommendation techniques as it is insensitive to the product details. However, a risk of potential privacy leaks exist in the recommendation process. For example, continual observation of the recommendations with some background information enables an adversary to infer the individuals rating or purchase history [96].

McSherry et al. [97] introduced a differential privacy mechanism to traditional recommender algorithms, adding Laplace noise into the covariance matrix. Zhu et al. [98] proposed a private k nearest neighbor (KNN) algorithm for neighborhood-based recommender systems while minimizing the accuracy loss of the recommendations. The algorithm can resist attacks on collaborative filtering and has the potential to be extended to other scenarios that need to

TABLE 16  
Three Aspects of the Data Lifecycle in Local Differential Privacy

| Original data type                        | Uploaded data type                                                       | Post-process target                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| single numeric or categorical attribute   | noisy attribute value                                                    | value estimation [108]; distribution estimation [109]                                                                                              |
| multiple numeric or categorical attribute | noisy attributes values                                                  | value estimation [108]; mean and frequency estimation, complex machine learning tasks [110]; multi-dimensional joint distribution estimation [111] |
| set-valued data                           | a random bit of noisy set-valued data [112]; noisy set-valued data [113] | frequent item set estimation [112]; discrete distribution estimation [113]                                                                         |
| encoding data                             | a random bit of the encoding of location                                 | count estimation [114]                                                                                                                             |

perform top-k private selection from candidates, such as top-k queries. Friedman et al. [99] presented a differentially private SVD algorithm to deal with the privacy problem in matrix factorization based recommender systems.

The majority of work on private recommender systems assume attackers are either public users or recommender servers. Liu et al. [100] proposed a hybrid approach which combines differential privacy with randomized perturbation not only to hide users private data from servers but also prevent privacy inference from public users. In practice, randomized perturbation can sometime improve the accuracy of the recommendations.

### 5.3 Differential Privacy in Genetic Data

Given the advances in genome sequencing technology, highly detailed genetic data are being generated inexpensively at exponential rates. The collection and analysis of these data may accelerate biomedical discoveries and support various applications, including personalized medical services. Despite all the benefits, Naveed et al. [101] reviewed the mitigation strategies for a wide variety of attacks and contextualized them from the perspective of medicine and public policy.

Several initial genome-wide association studies (GWAS) have explored the potential of differentially private data publishing by shifting the original location of the variants. Johnson et al. [102] developed a framework for *ab initio* exploration of case-control GWAS. This framework provides privacy-preserving answers to some key GWAS queries. They designed the operators to output the differentially private number of SNPs associated with the disease and other related statistical results. By considering protection against set membership disclosure, Tramer et al. [103] proposed a relaxation of the adversarial model of differential privacy and showed that this weaker setting achieves higher utility.

## 6 FUTURE DIRECTIONS

### 6.1 Adaptive Data Analysis

Adaptive data analysis creates a connection between differential privacy and machine learning. Machine learning theory is well developed, but is based on the assumption that a learning algorithm operates on a freshly sampled dataset. However, data samples are often reused, and the practice of learning is naturally adaptive. This adaptivity violates standard generalization guarantees. It is easy to over-fit the data when applying adaptive data analysis. Differential privacy can solve this adaptive problem.

Dwork et al. [104], [105] proposed a method that can perform arbitrary adaptive data with solid generalization

guarantees. In this method, the algorithm's stability is assumed to prevent over-fitting and differential privacy is considered to be an algorithmic stability guarantee. Thus, a quantitative error bound is provided in terms of generality.

This line of research proves that differential privacy has a generalization property that can be applied in statistical analysis and machine learning [106], [107]. Although, at this early stage, many new possibilities for generalization need to be explored.

### 6.2 Local Differential Privacy

The majority of work on differential privacy assumes that the curator is trustworthy; however, as more organizations have resorted to distributed systems for data acquisition, storage, and analysis, it has been noted that curators are less reliable in this distributed context.

Recently, a Local Differential Privacy (LDP) model has been proposed to address the issue. Noise is added on the user side before submitting the results to an un-trusted data curator. The curator then conducts post-processing on them to obtain acceptable statistics, which can be further shared with the public. Here in the statistics types, which are available for the data curator to obtain would be limited to the population statistics and depend on both the design of the local perturbation mechanism and the post-processing mechanism.

The local privacy model was first formalized in [45], and then a theoretic upper bound under the LDP model was provided by Duchi et al. [115]. Typically two main research questions have been investigated:

- 1) How to design acceptable LDP mechanisms for different original data types generated by distributed users?
- 2) How to design acceptable LDP mechanisms that can achieve different analysis targets such as value estimation, distribution estimation and more complex machine learning tasks?

In the past few years, related LDP works have focused on three aspects: the existing data types in user nodes (original data type), the data types submitted to data curators by users (uploaded data type), and the targets to achieve by analyzing the uploaded data (post-process target). Table 16 gives an overview of these three data types.

### 6.3 Differential Privacy for a Coupled Information

Existing research assumes that records are sampled independently from a data universe. However, in real-world applications, records are rarely independent. The

relationships among records are referred to as coupled information. A differential privacy technique performed on coupled information will disclose more information than expected, and this indicates a serious privacy violation [116], [117].

To deal with this problem, Kifer et al. proposed two privacy frameworks, *Pufferfish* [118] and *Blowfish* [117], to allow application domain experts to add an extra data relationship to develop a customized privacy definition. Based on *Pufferfish*, Yang et al. [119] proposed Bayesian differential privacy to deal with correlation problems. Wang et al. [120] observed that *non-independent reasoning* (NIR) allows the information about one record in the data to be learned from the information of other records in the data. Chen et al. [121] considered social networks and dealt with the coupled problem by multiplying the sensitivity with the number of coupled records. Zhu et al. [122] proposed a coupled sensitivity and designed a coupled data release mechanism.

Generally, when addressing privacy issues in a coupled dataset, the key problem is identifying which information is coupled, then modeling the extra background information introduced by coupled records. These are still open issues that need to be explored further.

#### 6.4 Differential Privacy and Mechanism Design

Mechanism design is a kind of game in which inputs are delivered by multiple agents. However, agents may manipulate their inputs, so the mechanisms design should ask agents to provide true inputs [123].

McSherry [11] first proposed a design for auction mechanisms using differentially private mechanisms as the building blocks. This private mechanism is only approximately truthful. Later, Nissim et al. [124] converted these private mechanisms into fully truthful mechanisms. Cummings et al. [125] studied the multi-dimensional aggregative games and solved the equilibrium selection problem. Barthe et al. [123] introduced a relational refinement system for verifying a mechanism's design in terms of differential privacy.

This series of research focuses on mechanism design based on differential privacy. They are good representatives for the interdisciplinary area of differential privacy and game theory.

#### 6.5 Relaxation of Differential Privacy

Differential privacy is indeed an excellent definition to guarantee strict privacy levels. The most attractive feature is that the privacy level degrades smoothly under composition theory. As pure differential privacy is too strict in practical terms, a relaxation of differential privacy is highly desired. Approximate or  $(\epsilon, \delta)$ -differential privacy is widely used as a method of relaxation and has been discussed heavily in this paper. However, the bound analysis for the composition of approximate differential privacy is hard [126], even for simple computations, making it inconvenient for use in real-world applications.

Dwork et al. [127] developed a new relaxation technique called concentrated differential privacy, in which the privacy loss in multiple computations has a small mean and is sub-Gaussian. Based on this new technique, Bun [128] proposed zero-concentrated differential privacy to minimize the mean of privacy losses to zero, to improve the bound of  $\delta$ . These relaxation definitions need to be further explored. For example, designing randomized algorithms to achieve the new

definitions and analyzing the sample complexity of those algorithms still remain problems. Generally, previous work on differential privacy can be re-visited through the perspective of new definitions, which may lead to new insights and results.

## 7 CONCLUSIONS

This paper presents a multi-disciplinary survey of work on differential privacy including an overview of the huge amount of literature in two major differential privacy research streams: data publishing and data analysis. We identified different publishing mechanisms for data publishing and compared various types of input and output data. In addition, we presented two basic dataset publishing methods: anonymized and learning-based. We discussed two basic frameworks for data analysis and illustrated their respective analysis scenarios. The basic techniques in differential privacy look simple and intuitively appealing, and when combined with specific problems, differential privacy demonstrates itself as a powerful and useful tool for a diverse range of applications.

Differential privacy still has much unknown potential, and the summarized literature in this paper is intended as a starting point for exploring new challenges in the future. Our goal is to provide an overview of existing works on differential privacy to show their use to newcomers, as well as experienced practitioners, in various fields. We also hope this review will help to prevent redundant or ad hoc efforts for both researchers and industry.

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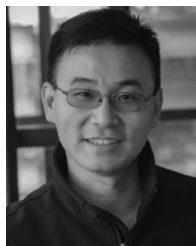
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